Important (Board & JEE)

Equipotential Surface :

The three dimensional surface in an electric field, having all points on it at same potential, is called an equipotential surface.

Properties:

(1) Since all point on an equipotential surface are at same potential, so no work is done in transferring a test charge along this surface.

[reason if $V_A = V_B$, $V_A - V_B = 0$

then $W_{ext} = q_{test} \times (V_A - V_B)$

from $B \to A$ $= 0$]

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V. V. Imp (2) Since no work is done by external agent or electric force. [$W_{ext} = -W_{el}$] in displacing a test charge on equipotential surface.

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it implies that electric force on test charge q_0 i.e. $q_0\vec{E}$ will always be perpendicular to displacement $\vec{\Delta r}$ of test charge q_0 on this surface.

$$W_{el} = \vec{F_{el}} \cdot \vec{\Delta r} = q_0 \vec{E} \cdot \vec{\Delta r}$$

$$\because W_{el} = 0 \quad \therefore q_0 \vec{E} \cdot \vec{\Delta r} = 0$$

$$\text{or } \vec{E} \cdot \vec{\Delta r} = 0 \Rightarrow \vec{E} \perp \vec{\Delta r}$$

electric field & field lines are always $\perp$ to equipotential surface

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Alternative

We know

$$dv = -\vec{E}\cdot d\vec{r}$$

For displacement $d\vec{r}$ along equipotential surface $dv = 0$ i.e. $\vec{E}\cdot d\vec{r} = 0$

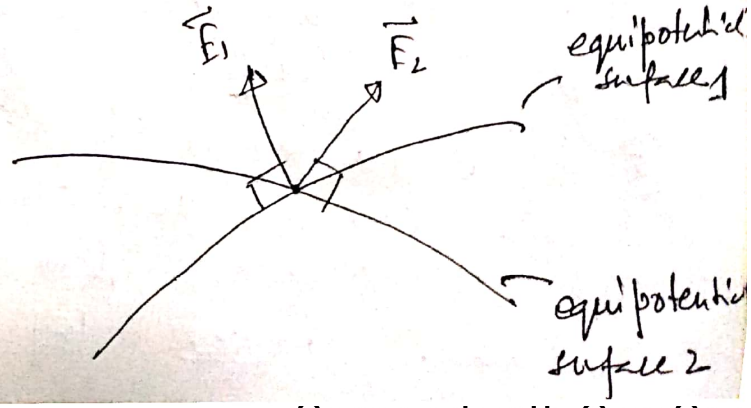
$$\therefore \vec{E} \perp d\vec{r}$$

i.e. electric field / electric line of force is always $\perp$ to equipotential surface.

(3) V.V.Imp Two equipotential surfaces can never intersect each other because otherwise the point of intersection will have two potentials which of course not acceptable.

Alternative :

We know electric field is perpendicular to equipotential surface. If two equipotential surface intersect, then electric field at the point of intersection will have two directions which is impossible



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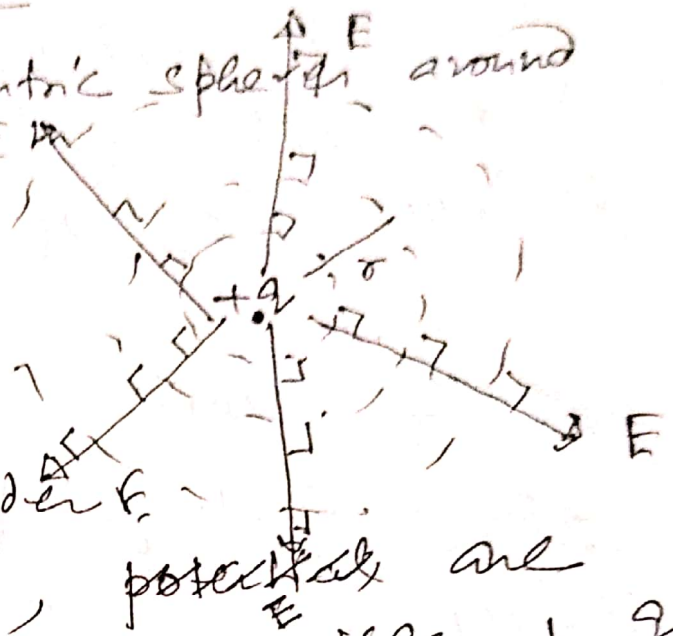
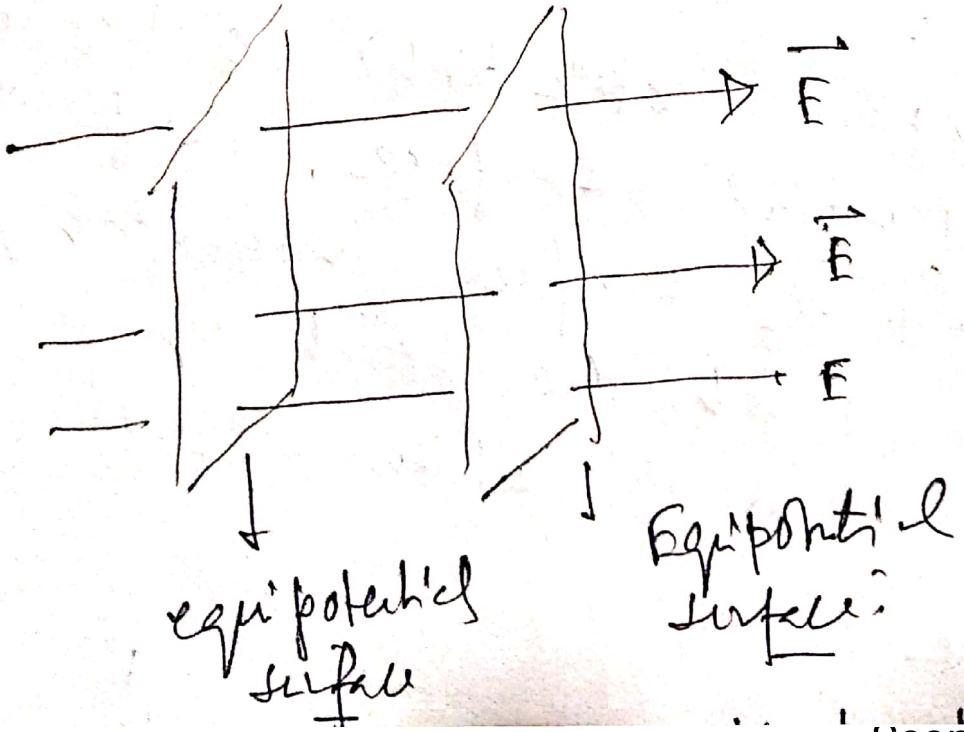
(4) example of equipotential surfaces.

(i) A series of concentric spheres around a point charge.



[reason: all pts on the surface of any one of these spheres, consider one of radius 'r', potential are at same potential $V = \frac{kq}{r} = \frac{1}{4\pi\epsilon_0}\frac{q}{r}$]

(ii) for uniform electric field, represented by straight, parallel equispaced lines of force, equipotential surfaces are ∥ planes ⊥ to electric line of force or electric field.



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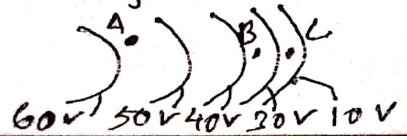
(5) for JEE In a family of equipotential surfaces, the surfaces are closer where electric field is stronger & farther apart where field is weaker. Reason: $|E| = \left|\frac{dv}{dr}\right|$, for same change in potential dv, $|E| \propto \frac{1}{|dr|}$ ie gap $|dr|$ betn equipotential surface is less where E is strong & viceversa

ex:



in example $E_C > E_B > E_A$.

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